

# DAY-2 — Networking Basics

CyberNotes • Detailed Beginner Reference (Improved Edition)

**Learning goal:** Build a practical, working understanding of **networking basics**, connect the concept to real cybersecurity work, and practice it safely in a local lab.

**Study rule:** Understand a command or concept before copying it. Security tools are powerful because they interact with real systems — your practice environment should always be intentionally controlled and authorized.

## 1. Core Concepts

**Core idea:** Networking Basics is a building block for practical cybersecurity. This section explains what each term actually means, where you'll encounter it, and how it connects to the bigger picture.

**Why it matters:** Security work depends on understanding normal system and network behavior before you can recognize abnormal behavior. This topic connects directly to Linux, networking, web security, monitoring, and troubleshooting — the threads running through this whole 20-day series.

**Vocabulary covered:** IP addresses, MAC addresses, ports, protocols, TCP/UDP, LAN/WAN, routing, subnet basics.

| Reference                            | What to remember              |
|--------------------------------------|-------------------------------|
| <code>ip addr</code>                 | show interfaces and addresses |
| <code>ip route</code>                | show routing table            |
| <code>ping &lt;host&gt;</code>       | test reachability             |
| <code>ss -tuln</code>                | show listening sockets        |
| <code>ip neigh</code>                | show neighbor table           |
| <code>traceroute &lt;host&gt;</code> | observe route hops            |

## 2. Concept Explanations

*Each term below is explained in plain language, with a note on why it matters for security work. Read these before running any commands.*

### 1. IP addresses

An IP address is a numeric label (like 192.168.1.10 for IPv4) that identifies a device on a network so traffic knows where to go. Private ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) are used inside local networks and aren't directly reachable from the internet, while public IPs are globally routable. Recognizing which range an address falls in tells you a lot about where a device sits and how exposed it is.

### 2. MAC addresses

A MAC address is a hardware identifier burned into a network interface (e.g. 00:1A:2B:3C:4D:5E), used for delivery within a local network segment (Layer 2). Unlike IP addresses, MACs normally don't change and

aren't routable across the internet, which is why attacks like ARP spoofing exploit the local trust in MAC-to-IP mappings. ``ip neigh`` shows the MAC-to-IP mapping table your machine currently knows about.

### 3. ports

A port is a numbered endpoint (0-65535) that lets a single IP address run many services at once — port 22 for SSH, 80 for HTTP, 443 for HTTPS. Scanning which ports are open on a host reveals what services are running and is one of the very first steps in both attacking and defending a system. ``ss -tuln`` shows which ports are actively listening on your own machine.

### 4. protocols

A protocol is an agreed-upon set of rules for how devices communicate — TCP, UDP, HTTP, and DNS are all protocols operating at different layers. Understanding a protocol's normal behavior is what lets you spot abnormal behavior, like a DNS query going to an unexpected server or an HTTP request missing expected headers.

### 5. TCP/UDP

TCP is connection-oriented: it performs a handshake, guarantees delivery, and is used for things like web browsing and file transfer. UDP is connectionless and faster but doesn't guarantee delivery, so it's used for things like DNS lookups and video streaming where speed matters more than perfect reliability. Many scanning and troubleshooting tools let you target TCP or UDP separately because they behave very differently under the hood.

### 6. LAN/WAN

A LAN (Local Area Network) is a network confined to one physical location, like a home or office, while a WAN (Wide Area Network) spans larger distances and connects multiple LANs together — the internet is the largest WAN. Security posture differs sharply between the two: LANs often trust internal traffic more, which is exactly what attackers exploit once they get a foothold inside one.

### 7. routing

Routing is the process of forwarding traffic between different networks based on destination IP address, using a routing table that says "to reach network X, send traffic via gateway Y." ``ip route`` shows your machine's routing table, and misconfigured routes are a very common cause of "I can reach some things but not others" networking problems.

### 8. subnet basics

A subnet divides a larger network into smaller, more manageable segments, expressed with a mask like /24 (255.255.255.0), which tells you how many addresses belong to that segment. Subnetting is used both for organization (separating departments or environments) and for security (isolating sensitive systems so a breach in one segment doesn't automatically expose everything else).

### 3. Safe Practical Lab

**Step 1.** Define the lab target and the exact question you are trying to answer.

**Step 2.** Run the smallest safe observation or command that can answer that question.

**Step 3.** Read the output carefully; record important fields, errors, timestamps, and assumptions.

**Step 4.** Change one variable at a time and compare the result.

**Step 5.** Write a short conclusion and clean up temporary files or services.

#### Example lab sequence:

```
ip addr
ip route
ss -tuln
ping -c 3 192.0.2.10
```

### 4. Common Mistakes & Troubleshooting

- Running a command without checking its target, current directory, or privileges.
- Confusing a tool's output with proof of a vulnerability or compromise — output is evidence, not a conclusion.
- Ignoring error messages instead of reading the exact reason for failure.
- Changing several settings at once, making it hard to tell what actually fixed or broke the lab.
- Saving real credentials, tokens, or personal data in practice notes or public repositories.
- Testing a public target just because it's reachable — reachability is never the same as authorization.

#### Troubleshooting checklist:

- What system am I on?
- What exact command/request did I run?
- What did I expect?
- What happened instead?
- What does the error message say?
- Can I reproduce it in a clean local lab?
- What changed since the last successful attempt?

## 5. Deep-Dive Questions

*Answer these in your own words after finishing the lab — they're designed to check real understanding, not memorization.*

**Q1.** What problem does IP addresses solve, in your own words?

Notes: \_\_\_\_\_

**Q2.** What is the difference between normal behavior and a suspicious observation in this topic?

Notes: \_\_\_\_\_

**Q3.** Which command, log, packet, file, or response would you inspect first, and why?

Notes: \_\_\_\_\_

**Q4.** What could cause a misleading result or false positive here?

Notes: \_\_\_\_\_

**Q5.** How would you document your observation so another learner could reproduce it safely?

Notes: \_\_\_\_\_

**Q6.** Which security control reduces the risk associated with this topic?

Notes: \_\_\_\_\_

## 6. Revision Checklist

- I can explain networking basics in my own words.
- I can identify the main terms without looking at the notes.
- I can perform the example in a controlled lab.
- I can interpret the output instead of only copying it.
- I know at least one defensive or troubleshooting use for the concept.
- I can document the result clearly and safely.

## 7. Safety & Scope

Use these commands and tools only on systems you own, a local virtual machine, a CTF, an intentionally vulnerable application, or a target for which you have explicit written permission.

Keep a written scope: target, allowed actions, time window, and stop conditions. Never scan or test random public systems.

When practicing, save your notes and outputs so you can explain what you observed rather than blindly running commands.

**Next step:** After Day-2, review your notes and connect this topic to the next subject in the series. The goal is not to memorize commands — it's to understand systems well enough to reason about security.

## 8. Quick Reference & Personal Notes

### Topic: Day-2 — Networking Basics

Three things I learned:

1. \_\_\_\_\_
1. \_\_\_\_\_
1. \_\_\_\_\_

### Commands / concepts to memorize:

- `ip addr` — show interfaces and addresses
- `ip route` — show routing table
- `ping` — test reachability
- `ss -tln` — show listening sockets
- `ip neigh` — show neighbor table
- `traceroute` — observe route hops

### My lab result

What I tested: \_\_\_\_\_

What I observed: \_\_\_\_\_

What I would investigate next: \_\_\_\_\_

**Final reminder:** A strong cybersecurity learner can explain both the offensive concept and the defensive reason for learning it. Keep practice authorized, reproducible, and documented.